

TH-02: Frame-Level Identifiability of HM3 on $\text{St}(d, r)$

Score-design line, slot-2 / rank-r decision

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1. Problem statement

Let $A \in \mathbb{R}^{N \times d}$ be the underlying matrix whose top- r right singular subspace $V_r := V_{\text{exact}}[:, :r]$ we are trying to recover online. At a given streaming block we have access to three row-blocks of A :

- $A_{\text{sketch}} = s \cdot V_{\text{state}}^\top \in \mathbb{R}^{r \times d}$, the rank- r carry of the prefix;
- $A_{\text{cur}} \in \mathbb{R}^{m \times d}$, the half-window of new rows;
- $A_{\text{fut}} \in \mathbb{R}^{m \times d}$, the peek half-window after that.

Define the magnitude-only frame energies

$$u_X(V) = \frac{\|A_X V\|_F^2}{\|A_X\|_F^2}, \quad X \in \{\text{sketch}, g_1, g_2\},$$

and the rank- r frame score

$$Z(V) = \text{HM3}(u_{\text{sk}}(V), u_{g_1}(V), u_{g_2}(V)) = \frac{3}{u_{\text{sk}}(V)^{-1} + u_{g_1}(V)^{-1} + u_{g_2}(V)^{-1}}, \quad V \in \text{St}(d, r).$$

Let $\Pi := \Pi_{\text{row}(B_\cup)}$ denote orthogonal projection onto the union row span $B_\cup = \text{row}(A_{\text{sketch}}) \cup \text{row}(A_{\text{cur}}) \cup \text{row}(A_{\text{fut}})$.

The question. Characterise the conditions under which

$$\arg \max_{V \in \text{St}(d, r)} Z(V) = \Pi V_r \tag{*}$$

holds, or construct an explicit counterexample family showing (*) fails for any reasonable A1-style assumption on $(A_{\text{sketch}}, A_{\text{cur}}, A_{\text{fut}})$.

Why this matters. The empirical kill/continue gate (FAM-01-DIAG) tests whether $Z_{\text{oracle}} \geq Z_{\text{winner}}$ on a handful of §6 matrices and blocks. A theoretical characterisation says *when* it must hold (or when it cannot), and this is what decides the score-design path at the limit. The two gates are complementary, not redundant.

2. Three target deliverables

A landed result on $(\star)$ takes one of the following forms; each changes the next action of the project.

(D1) Positive theorem. A statement of the form

Under (A1), a spectral-spread condition $C(\Sigma, r, m)$, and mild moment regularity on the row distribution, $(\star)$ holds with $Z_{\text{oracle}} - \sup_{V \perp \Pi V_r} Z(V) \geq \delta(C)$.

Then C becomes a value-only certifiable side condition: a window in which C holds gets the score-family ship; a window in which C fails falls back to iSVD. This is the missing TH-02 (HM3 as a population lower bound), upgraded to argmax-correctness on $\text{St}(d, r)$.

(D2) Counterexample family. A parametric construction of $(A_{\text{sketch}}, A_{\text{cur}}, A_{\text{fut}})$ on which

$$\sup_{V \in \text{St}(d, r)} Z(V) > Z(\Pi V_r) + \eta$$

for some $\eta > 0$, parametrised by $(d, r, m, \text{tail shape})$, such that the construction is not pathological (it satisfies A1 and a row-population assumption that holds on §6 matrices). This is Q12 sharpened to rank- r frames, and it kills the entire magnitude-only path *permanently* (not just for the current suite). The pivot to Q13 / Q14 evidence features becomes the only remaining option.

(D3) Sharp boundary. A separation showing magnitude-only frame scores cannot identify V_r when the visible-window covariance $\widehat{\Sigma}_w$ has sub-leading eigen-gap below threshold $\delta(\Sigma, m, r)$. This matches the empirical **mixed-tail-sharp** pathology and motivates cross-window consistency (Q13) as the load-bearing reliability feature, with a quantified condition for when it is needed.

3. Block-1 noise-free reduction (entry point)

The cleanest sub-question, in which $A_{\text{sketch}} = 0$ so HM3 reduces to the two-term harmonic mean $\text{HM2}(u_{g_1}, u_{g_2})$. Consider an i.i.d. row model

$$A_{\text{cur}}, A_{\text{fut}} \stackrel{\text{iid}}{\sim} P_{\Sigma}, \quad \mathbb{E}_{a \sim P}[aa^\top] = \Sigma, \quad \text{rank}(\Sigma) \geq r,$$

and ask:

Does $\arg \max_{V \in \text{St}(d, r)} \text{HM2}(u_{g_1}(V), u_{g_2}(V)) \rightarrow V_r$ in probability as $m \rightarrow \infty$, and what is the finite- m leak?

What is already known empirically. On **diffuse-diffuse** block 1, the optimizer of S6 (which reduces to HM2 under no-carry) recovers $V_{\text{exact}}[:, 1]$ at $\cos^2 \approx 0.999$ and $V_{\text{exact}}[:, 2]$ at $\cos^2 \approx 0.982$ (overview §1bis, M3 verification); on the same block of **mixed-tail-sharp**, DIAG-05 records a 4% HM-evi gap in favour of a non-oracle direction. The theoretical question is whether the 4% is

- a finite-sample artifact that vanishes as $m \rightarrow \infty$ (in which case the population-level HM2 identifies V_r and only finite- m stability matters — direction (D1)), or
- a structural identifiability failure of the magnitude-only population score on the heavy-tail family (in which case it persists at every m , and the construction generalises to (D2)).

Either answer is informative, and the question is one paragraph wide.

A natural ansatz for (D2). Take P_Σ to be a mixture

$$P_\Sigma = (1 - \varepsilon)\mathcal{N}(0, \Sigma_{\text{bulk}}) + \varepsilon\mathcal{N}(0, \Sigma_{\text{tail}}),$$

with Σ_{tail} rank-deficient on $\text{span}(V_r)^\perp$ and supported on a direction $v_{\text{conspiracy}}$ that has high response in both windows by correlated tail events but vanishing population correlation with V_r . The empirical §1ter “tail-conspiracy” framing is exactly this construction; what is missing is a closed-form proof that $Z(v_{\text{conspiracy}}) > Z(V_r)$ at the population level for some $\varepsilon, \Sigma_{\text{tail}}$ choice.

4. Why this is the highest-value theoretical problem now

- It is the limit-case version of the empirical kill/continue gate. **FAM-01-DIAG** answers “does HM3 identify the oracle frame on *these* matrices and blocks?” (★) answers “under what conditions *must* it?” The empirical and theoretical gates jointly decide the score-design path: a positive theory result + a passing empirical gate justifies shipping with a value-only certifiable side condition; an impossibility result + a failing empirical gate kills the family permanently rather than matrix-by-matrix.
- It subsumes three currently open / underpowered theoretical items: Q12 (scalar-score impossibility), TH-02 (HM3 as a population lower bound), and FB-006 (the underpowered codim-1 plateau argument). The plateau geometry is a corollary of (★)’s failure mode, not an independent statement.
- Q9 (entropy proxy theory) and TH-01 (HM3 max-min stability surrogate) are already correctly bounded in the docs and do not change the next action. Tail-conspiracy formalisation (§1ter) is informal and depends on (★) to land.

- The block-1 reduction is a one-paragraph question whose answer either way is publishable and decision-relevant.

5. Success criteria

A landed result must include:

1. A precise statement of $(\star)$ at the population level ($A_{\text{cur}}, A_{\text{fut}}$ from a stated row distribution), not just at the empirical-row level. The score family is an *inductive bias*, not an estimator of $\|Av\|^2$; identifiability must be stated for the bias’s intended population target.
2. Either: (D1) a sufficient condition C that holds on at least 3 of `{mixed-tail-soft, diffuse-diffuse, etf-basket-basis}` and fails on `{mixed-tail-sharp, static-cex}`, with a value-only test of C ; or (D2) a counterexample family parametrised by interpretable quantities (ε , tail rank, eigengap), with explicit $Z(V_{\text{cex}}) - Z(V_r) > \eta(\varepsilon)$ and a numerical verification on a synthetic instance.
3. A statement of the relationship to TH-01 (max-min surrogate) and to the rank- r Stiefel lift: $(\star)$ at $r = 1$ vs. $r \geq 2$ should differ in a way that explains the empirical `etf-basket-basis` gap ($S6 \cos 1^2 = 0.807$ vs. $\text{online}_{\text{VO}} \cos 1^2 = 0.969$, a -0.162 gap that is the only live operational target after FB-001).

6. Out of scope

- Optimizer convergence on $\text{St}(d, r)$ (this is INFRA-02 / FAM-01 implementation, not a theoretical identifiability statement).
- Per-row leverage-aware weightings (Q4) — bias-axis-orthogonal to identifiability and only matters after $(\star)$ is settled.
- Sample-complexity bounds for fixed δ — nice to have but the qualitative identifiability question is more decision-relevant first.

7. Relation to current backlog

- Resolves: Q12 (scalar-score impossibility), TH-02 (population lower bound), FB-006 (plateau-dimension motivation).
- Complements: FB-004 / FAM-01-DIAG (empirical gate), Q13 (cross-window consistency, the natural follow-up if (D2) lands).
- Independent of: FB-002, FB-003, FB-005 (documentation/data rigor); FAM-02 (row-concentration regime).